IDX Smart Money Paper Trading System Blueprint

Objective
Build a modular system to detect institutional accumulation in Indonesian equities, simulate trades, backtest strategies, and later support live execution.

Architecture
Data ingestion -> Database -> Feature engineering -> Smart Money Score -> Strategy engine -> Backtester -> Paper trading -> SwiftUI dashboard.

Data
OHLCV, tick data, order book, broker summary, foreign flow, corporate actions, financial statements, news sentiment.

Database
Tables: stocks, prices, brokers, foreign_flow, corporate_actions, signals, trades, portfolios, backtests.

Signal Engine
Compute volume spike, RVOL, VWAP, trend, broker accumulation, foreign accumulation, order-book imbalance, volatility.

Smart Money Score
Weighted score (100 points) combining all signals. Tune weights through historical optimization.

Trading Rules
Daily scan; buy highest scores meeting liquidity and trend filters; exit by stop loss, target, trailing stop, or score deterioration.

Risk
Max portfolio exposure, position sizing, sector limits, drawdown control.

Backtesting
Walk-forward testing, transaction costs, slippage, multiple market regimes, performance metrics.

ML Roadmap
Predict forward returns using engineered features with XGBoost/LightGBM; compare against rule-based system.

Tech Stack
Python (pandas, polars, vectorbt/backtrader, FastAPI, PostgreSQL), Redis, SwiftUI frontend.

Claude Tasks
Implement modules incrementally with unit tests, documentation, and reproducible backtests.